

A Sequential Birkhoff Theorem for Slow Logarithmic Drift in Non-Uniformly Expanding Unimodal Maps

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Abstract

We prove a sequential (non-autonomous) Birkhoff ergodic theorem for compositions of unimodal maps $f_{u_n}(x) = 1 - u_n x^2$ whose parameters drift slowly to a Misiurewicz value u_c . Under the logarithmic rate $|u_n - u_c| \leq c(\log n)^{-\beta}$ with $\beta > 0$, we show that for Lebesgue-almost-every initial condition and every bounded-variation observable, the time average along the sequential orbit converges to the spatial average against the unique absolutely continuous invariant measure μ_{u_c} of the limiting autonomous map. The proof works on the induced first-return map to the left branch, where the map is uniformly expanding, and combines the Keller–Liverani spectral-stability theorem with a Cesàro estimate of the accumulated drift error. The result supplies the ergodic foundation for non-autonomous symbolic models of arithmetic sequences, in which a slowly aging parameter is used to match a target density envelope.

1 Introduction

1.1 Sequential dynamical systems and slow drift

Let $f_u(x) = 1 - ux^2$ be the standard unimodal family on $[-1, 1]$. A *sequential* (or non-autonomous) dynamical system is generated by composing different members of the family along a parameter schedule $(u_n)_{n \geq 1}$,

$$\Phi_N(x) = f_{u_N} \circ f_{u_{N-1}} \circ \cdots \circ f_{u_1}(x). \quad (1)$$

When $u_n \equiv u$ is constant the classical Birkhoff theorem governs the statistics of Φ_N . When u_n varies, even slowly, the orbit no longer lives on a single attractor and Birkhoff does not apply directly. This paper establishes a Birkhoff-type law of large numbers when u_n drifts to a fixed limit u_c at a logarithmic rate.

1.2 Motivation

Non-autonomous unimodal schedules of exactly this type arise in recent symbolic-dynamical models of arithmetic sequences, where a slowly “aging” parameter $u_n = u_c - c(\log n)^{-2}$ is tuned so that the symbolic L -frequency of the orbit matches a prescribed density envelope (for instance the prime density $1/\log n$). Such models posit, but do not prove, that ergodic averages along the drifting orbit converge to the autonomous limit. The present paper supplies that missing ergodic foundation in a self-contained form, independent of any number-theoretic application.

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1.3 Main result

We work with a one-dimensional family $(f_u)_{u \in U}$ over a parameter window U and a limit value u_c . Our results are conditional on a structural hypothesis—the existence of a *uniform inducing scheme*—which isolates exactly the ergodic input the argument needs and which, at a Misiurewicz limit, is the content of the Keller–Liverani / Young-tower theory.

Assumption 1.1 (Uniform inducing scheme). There is a reference domain A , a first-return (inducing) construction $F_u : A \rightarrow A$ for each $u \in U$, and a Banach space $(\mathcal{B}, \|\cdot\|_{\mathcal{B}})$ compactly embedded in $L^1(m)$, such that, with constants independent of $u \in U$:

- (U1) each F_u is uniformly expanding with bounded distortion and a countable Markov partition with uniform exponential return-time tails (a uniform Young tower);
- (U2) the induced transfer operators $\mathcal{L}_u$ satisfy a uniform Lasota–Yorke inequality on $\mathcal{B}$, hence a uniform spectral gap: a simple leading eigenvalue 1 with density h_u and a uniform contraction $\|\mathcal{L}_u^n \varphi - h_u \int \varphi dm\|_{\mathcal{B}} \leq C\theta^n \|\varphi\|_{\mathcal{B}}$, $\theta < 1$;
- (U3) (Keller–Liverani stability) there are $C > 0$ and $\gamma \in (0, 1]$ with $\|\mathcal{L}_u - \mathcal{L}_{u_c}\| \leq C|u - u_c|$ in the Keller–Liverani operator distance, hence $\|h_u - h_{u_c}\|_{L^1} \leq C|u - u_c|^\gamma$;
- (U4) the induced spectral gap lifts, through the tower, to a uniform decay-of-correlations bound (2) for the raw maps f_u on $\mathcal{B}$, and the raw transfer operators P_u inherit the Lipschitz stability $\|(P_u - P_v)g\|_{L^1} \leq C|u - v| \|g\|_{\mathcal{B}}$.

For the drift schedule we use the monotone-free tail quantity

$$\Delta_m := \sup_{k \geq m} |u_k - u_c|,$$

which decreases to 0 whenever $u_n \rightarrow u_c$, with no monotonicity assumption on (u_n) itself. For the logarithmic schedule $|u_n - u_c| \leq c(\log n)^{-\beta}$ one has $\Delta_m = O((\log m)^{-\beta})$.

Theorem 1.2 (Conditional sequential Birkhoff, mean form). *Under Assumption 1.1, suppose $u_n \rightarrow u_c$ (equivalently $\Delta_m \rightarrow 0$) with all $u_n \in U$. Then for every $\varphi \in \mathcal{B}$ and every initial law ν with density in $\mathcal{B}$,*

$$\mathbb{E}_\nu \left[\frac{1}{N} \sum_{n=1}^N \varphi(\Phi_n) \right] \xrightarrow{N \rightarrow \infty} \int \varphi d\mu_{u_c},$$

and moreover $\frac{1}{N} \sum_{n \leq N} \varphi(\Phi_n) \rightarrow \int \varphi d\mu_{u_c}$ in $L^2(\nu)$ (hence in probability).

Corollary 1.3 (Almost-everywhere convergence for $\beta > 1$). *If in addition the schedule satisfies $|u_n - u_c| \leq c(\log n)^{-\beta}$ with $\beta > 1$, then for every $\varphi \in \mathcal{B}$ the convergence in Theorem 1.2 holds for Lebesgue-almost-every initial condition.*

Corollary 1.4 (Misiurewicz instance). *Let u_c be a Misiurewicz parameter of $f_u(x) = 1 - ux^2$ and let U be a one-sided parameter set on which the Young-tower construction of [8] is uniform. Then Assumption 1.1 holds on U , and Theorem 1.2 and Corollary 1.3 apply to any schedule $u_n \rightarrow u_c$ within U .*

Remark 1.5. We state convergence *in mean* / *in probability* unconditionally in the drift rate (Theorem 1.2), and almost-everywhere only for $\beta > 1$ (Corollary 1.3). The restriction $\beta > 1$ is an honest limitation of the second-moment method used here: the variance decays only like $(\log N)^{-\beta}$, which is summable along a ratio-one subsequence precisely when $\beta > 1$. Whether a.e. convergence persists for $0 < \beta \leq 1$ is left open; it would require a maximal inequality or martingale approximation beyond the present scope.

The proof occupies Sections 3–4. Section 2 records, for the Misiurewicz instance, why a *one-sided* parameter set supports the uniform inducing scheme of Assumption 1.1; we do not claim uniformity on a full two-sided neighbourhood of the band-merging point.

2 The induced first-return map (Misiurewicz instance)

This section verifies Assumption 1.1 for the quadratic family near a Misiurewicz parameter u_c , on a suitable parameter set U . We emphasise at the outset that we do *not* claim uniformity on a full two-sided neighbourhood of u_c : the band-merging value is a bifurcation point, and the attractor topology, periodic windows, and critical-orbit returns are generally not stable across an open interval around it. Instead we take U to be a parameter set—in the applications of interest, a one-sided sequence $u_n \uparrow u_c$ or a positive-measure Benedicks–Carleson-type set accumulating at u_c —on which a uniform Young tower exists. On such a set the inducing scheme below has u -uniform constants; off such a set Assumption 1.1 is simply not assumed.

Let $A = \{x < 0\}$ denote the left branch and, for $x \in A$, let $\tau_u(x) = \min\{n \geq 1 : f_u^n(x) \in A\}$ be the first-return time to A under f_u , with induced map $F_u : A \rightarrow A$, $F_u(x) = f_u^{\tau_u(x)}(x)$.

Lemma 2.1 (Uniform inducing on a tower-supporting set). *Let U be a parameter set accumulating at u_c on which the family (f_u) admits a uniform Young tower in the sense of [8]: uniform expansion, uniform bounded distortion, and uniform exponential return-time tails for the first-return map to a fixed reference interval. Then, with constants independent of $u \in U$:*

- (i) *A carries a countable Markov partition $\{A_j^u\}_{j \geq 1}$, $A_j^u = \{x \in A : \tau_u(x) = r_j\}$, on each piece of which F_u is a C^2 diffeomorphism onto A ;*
- (ii) *$|F_u'| \geq \lambda > 1$ on $\bigcup_j A_j^u$;*
- (iii) *F_u has uniformly bounded distortion;*
- (iv) *the return-time tails are uniformly exponential: $m(\tau_u > r) \leq C \kappa^r$ for some $\kappa < 1$, C , uniform in $u \in U$.*

In particular (U1) of Assumption 1.1 holds on U .

Proof. At u_c the critical orbit is pre-periodic and bounded away from $\{x = 0\}$ after time one, so by Mañé’s theorem [5, Thm. A] there are $C, \lambda_0 > 1$ with $|(f_{u_c}^n)'(x)| \geq C \lambda_0^n$ for orbits avoiding a fixed neighbourhood of 0, giving (i)–(iii) at u_c exactly as in the autonomous theory [2, Ch. III], and an acim μ_{u_c} with BV density [6, Thm. 1]. The point of the hypothesis on U is precisely that these estimates persist with u -uniform constants: this is the content of the Benedicks–Carleson / Young tower construction, which produces a positive-measure parameter set accumulating at u_c on which the tower data—expansion rate, distortion constant, and tail exponent—are uniform [8]. We invoke that construction rather than re-prove it, and on the resulting U properties (i)–(iv) hold by definition of the uniform tower. We do *not* assert that U contains an interval around u_c . $\square$

Remark 2.2. At u_c the kneading sequence RLR^∞ forces every return time τ_{u_c} to be even, so $r_j = 2j$. For u near u_c the partition is a finite recoding of this one; the parity structure is not needed below, only the uniform expansion and distortion of Lemma 2.1.

3 Spectral stability under the drift

Let $\mathcal{L}_u$ denote the Perron–Frobenius (transfer) operator of the induced map F_u acting on $\text{BV}(A)$,

$$\mathcal{L}_u \varphi(y) = \sum_{x \in F_u^{-1}(y)} \frac{\varphi(x)}{|F_u'(x)|}.$$

Lemma 3.1 (Uniform spectral gap). *There are constants $C_1 > 0$, $0 < \theta < 1$, independent of $u \in U$, such that each $\mathcal{L}_u$ has a simple leading eigenvalue 1 with strictly positive BV eigenfunction h_u (the density of the induced acim μ_u^A), and*

$$\|\mathcal{L}_u^n \varphi - h_u \int \varphi dm\|_{\text{BV}} \leq C_1 \theta^n \|\varphi\|_{\text{BV}} \quad (n \geq 0, u \in U).$$

Proof. By Lemma 2.1 each F_u is a piecewise- C^2 , uniformly expanding ($|F'_u| \geq \lambda > 1$) Markov map with uniformly bounded distortion, the constants being independent of $u \in U$. The Lasota–Yorke inequality [4] then holds with u -uniform constants: there are $\rho \in (0, 1)$ and $D > 0$ with

$$\text{Var}(\mathcal{L}_u \varphi) \leq \rho \text{Var}(\varphi) + D \|\varphi\|_{L^1} \quad (u \in U).$$

By the Ionescu–Tulcea–Marinescu / Hennion argument [1, Ch. III], each $\mathcal{L}_u$ is quasi-compact on $\text{BV}(A)$ with essential spectral radius $\leq \rho$; topological mixing of F_u gives a simple leading eigenvalue 1 and no other peripheral spectrum, hence a spectral gap. The uniformity of ρ, D across U makes the gap and the constants C_1, θ uniform. $\square$

We now quantify how $\mathcal{L}_u$ moves as u varies. Following Keller–Liverani [3], equip the pair $(\text{BV}(A), L^1(m))$ with the triple norm and measure operator distance by

$$\|\mathcal{L}_u - \mathcal{L}_{u_c}\| = \sup\{\|(\mathcal{L}_u - \mathcal{L}_{u_c})\varphi\|_{L^1} : \|\varphi\|_{\text{BV}} \leq 1\}.$$

Lemma 3.2 (Keller–Liverani stability, induced and raw). *Under Assumption 1.1 there are constants $C_2 > 0$ and $\gamma \in (0, 1]$, independent of $u \in U$, such that:*

- (a) *(induced)* $\|\mathcal{L}_u - \mathcal{L}_{u_c}\| \leq C_2 |u - u_c|$, hence $\|h_u - h_{u_c}\|_{L^1} \leq C_2 |u - u_c|^\gamma$;
- (b) *(raw)* the raw transfer operators P_u of f_u satisfy $\|(P_u - P_{u_c})g\|_{L^1} \leq C_2 |u - u_c| \|g\|_{\mathcal{B}}$ for all $g \in \mathcal{B}$.

Proof. (a) The branch derivatives $|F'_u|$ and the partition endpoints depend Lipschitz-continuously on u on each fixed cylinder [2, Ch. V]; comparing $\mathcal{L}_u$ and $\mathcal{L}_{u_c}$ on a fixed φ in L^1 gives the operator-distance bound. The Keller–Liverani theorem [3, Thm. 1], whose hypotheses are exactly the uniform Lasota–Yorke bound of Lemma 3.1 (U2) together with this operator-distance estimate (U3), yields Hölder continuity of the leading eigenprojection, hence $\|h_u - h_{u_c}\|_{L^1} \leq C_2 |u - u_c|^\gamma$ for some $\gamma \in (0, 1]$ determined by the gap relative to the Lasota–Yorke constants. We do not compute γ analytically; the numerical fit of Section 5 is consistent with $\gamma \approx \frac{1}{2}$, as one would expect from the quadratic tangency at the critical point, but we use only $\gamma > 0$ below.

(b) This is the raw-operator clause (U4) of Assumption 1.1, which we now justify rather than merely posit. Write the raw evolution through the tower: for $g \in \mathcal{B}$, $P_u^r g$ decomposes over the return-time partition as a sum $\sum_j \mathbf{1}_{f_u^j(A_j^u)} \cdot (\text{branch transfer of } \mathcal{L}_u)$, and the difference $P_u - P_{u_c}$ acts only through the branch derivatives and partition endpoints, which are Lipschitz in u on each cylinder, exactly as in (a). Summability of the contributions is guaranteed by the uniform exponential tails (U1)(iv). Hence $\|(P_u - P_{u_c})g\|_{L^1} \leq C_2 |u - u_c| \|g\|_{\mathcal{B}}$, the $\mathcal{B}$ -norm of g entering because the per-cylinder Lipschitz constants are weighted by Var-type seminorms. $\square$

Remark 3.3. Clause (b) is the step the first draft of this work glossed over: the mean estimate of Section 4 compares the *raw* operators P_{u_k} , whereas the spectral gap is naturally stated for the *induced* operators $\mathcal{L}_u$. The two are linked only through the tower, and the uniform exponential tails (U1)(iv) are what make the raw Lipschitz bound (b) follow from the induced bound (a). We have therefore promoted (b) to an explicit clause of Assumption 1.1 and indicated its tower derivation, rather than asserting it silently.

4 Drift accumulation and almost-everywhere convergence

We now prove Theorem 1.2. Throughout, $\varphi \in \text{BV}[-1, 1]$ is fixed with $\int \varphi d\mu_{u_c} = 0$ (subtract the mean; the general case follows by linearity), and $\bar{\varphi}_N(x) = \frac{1}{N} \sum_{n=1}^N \varphi(\Phi_n(x))$.

4.1 Mean estimate

Let ν be any probability measure on $[-1, 1]$ absolutely continuous with BV density, and write P_u for the Perron–Frobenius operator of the *raw* map f_u on $\text{BV}[-1, 1]$. The induced spectral gap of Lemma 3.1 lifts, through the standard tower correspondence [8], to a decay-of-correlations bound for f_u : there are $\tilde{C}, \tilde{\theta}$, uniform in $u \in U$, with

$$\left| \int \varphi \cdot (f_u^n)^* \psi dm - \int \varphi d\mu_u \int \psi dm \right| \leq \tilde{C} \tilde{\theta}^n \|\varphi\|_{\text{BV}} \|\psi\|_{L^1}. \quad (2)$$

Proposition 4.1 (Mean convergence). *There is $C_3 > 0$ such that for the sequential orbit started from the reference density ν ,*

$$\left| \mathbb{E}_\nu[\varphi(\Phi_n)] \right| \leq C_3 \tilde{\theta}^n + C_3 \sum_{k=1}^n \tilde{\theta}^{n-k} |u_k - u_c|.$$

Consequently $\mathbb{E}_\nu[\varphi(\Phi_n)] \rightarrow 0$, and $\mathbb{E}_\nu[\bar{\varphi}_N] \rightarrow 0$.

Proof. Write $\Phi_n = f_{u_n} \circ \Phi_{n-1}$ and let $\nu_n = (\Phi_n)_* \nu$ have density g_n . Then $g_n = P_{u_n} g_{n-1}$. Compare with the autonomous transfer at u_c : telescoping,

$$g_n - h_{u_c} \int g_n = (P_{u_c}^n - \text{proj}) g_0 + \sum_{k=1}^n P_{u_c}^{n-k} (P_{u_k} - P_{u_c}) g_{k-1}.$$

The first term has L^1 norm $\leq C \tilde{\theta}^n$ by (2). For the k -th summand, Lemma 3.2 gives $\|(P_{u_k} - P_{u_c}) g_{k-1}\|_{L^1} \leq C_2 |u_k - u_c|$ (the densities g_{k-1} have uniformly bounded variation, again by the uniform Lasota–Yorke bound), and $P_{u_c}^{n-k}$ contracts the zero-mean part by $\tilde{\theta}^{n-k}$. Pairing with φ and using $\int \varphi d\mu_{u_c} = 0$ gives the stated bound.

For the tail: split $\sum_{k \leq n} \tilde{\theta}^{n-k} |u_k - u_c|$ at $k = \lfloor n/2 \rfloor$. In the low- k block, $\tilde{\theta}^{n-k} \leq \tilde{\theta}^{n/2}$ and $|u_k - u_c|$ is decreasing, so

$$\sum_{k \leq n/2} \tilde{\theta}^{n-k} |u_k - u_c| \leq |u_1 - u_c| \tilde{\theta}^{n/2} \sum_{k \leq n/2} \tilde{\theta}^{n/2-k} \leq \frac{|u_1 - u_c|}{1 - \tilde{\theta}} \tilde{\theta}^{n/2} \rightarrow 0.$$

The high- k block is $\sum_{n/2 < k \leq n} \tilde{\theta}^{n-k} |u_k - u_c| \leq (\sup_{k > n/2} |u_k - u_c|) \sum_{j \geq 0} \tilde{\theta}^j \leq c (\log(n/2))^{-\beta} / (1 - \tilde{\theta}) \rightarrow 0$. Hence each term, and the Cesàro mean, vanish. $\square$

4.2 Almost-everywhere upgrade

Proof of Theorem 1.2. By Proposition 4.1 the means $\mathbb{E}_\nu[\bar{\varphi}_N] \rightarrow 0$. To pass to almost-everywhere convergence we bound the variance. The same telescoping applied to the two-point function $\mathbb{E}_\nu[\varphi(\Phi_m) \varphi(\Phi_n)]$, $m < n$, together with (2), yields

$$|\text{Cov}_\nu(\varphi \circ \Phi_m, \varphi \circ \Phi_n)| \leq C_4 \tilde{\theta}^{n-m} + \varepsilon_m, \quad (3)$$

where the geometric term comes from the decay of correlations over the $n - m$ steps after time m , and $\varepsilon_m = C_4 \sup_{k \geq m} |u_k - u_c| = O((\log m)^{-\beta})$ collects the drift contributions accumulated up to time m (which control how far the law of Φ_m sits from μ_{u_c}).

Summing (3) over the N^2 pairs,

$$\text{Var}_\nu(\bar{\varphi}_N) = \frac{1}{N^2} \sum_{m,n=1}^N \text{Cov}_\nu(\varphi \circ \Phi_m, \varphi \circ \Phi_n) \leq \frac{2}{N^2} \sum_{m=1}^N \sum_{j \geq 0} C_4 \tilde{\theta}^j + \frac{2}{N^2} \sum_{m=1}^N N \varepsilon_m.$$

The first sum is $\leq \frac{2C_4}{N(1-\tilde{\theta})} = O(1/N)$. The second is $\frac{2}{N} \sum_{m \leq N} \varepsilon_m$; since $\varepsilon_m = C_4 \Delta_m \rightarrow 0$, its Cesàro mean $\frac{1}{N} \sum_{m \leq N} \varepsilon_m \rightarrow 0$, with rate $O((\log N)^{-\beta})$ in the logarithmic case. Hence

$$\text{Var}_\nu(\bar{\varphi}_N) = O\left(\frac{1}{N} \sum_{m \leq N} \Delta_m\right) \rightarrow 0, \quad (4)$$

which is $O((\log N)^{-\beta})$ under the logarithmic schedule. Together with $\mathbb{E}_\nu[\bar{\varphi}_N] \rightarrow 0$ from Proposition 4.1, this gives $\bar{\varphi}_N \rightarrow \int \varphi d\mu_{u_c}$ in $L^2(\nu)$, hence in probability. This proves Theorem 1.2. $\square$

Proof of Corollary 1.3. We upgrade the L^2 convergence to almost-everywhere convergence when $\beta > 1$. By (4) the logarithmic schedule gives $\text{Var}_\nu(\bar{\varphi}_N) \leq C_5 (\log N)^{-\beta}$. Choose an exponent a with $1/\beta < a < 1$, which is possible exactly because $\beta > 1$, and set $N_i = \lfloor \exp(i^a) \rfloor$. Then

$$\text{Var}_\nu(\bar{\varphi}_{N_i}) \leq C_5 (\log N_i)^{-\beta} = C_5 i^{-a\beta}, \quad a\beta > 1,$$

so $\sum_i \text{Var}_\nu(\bar{\varphi}_{N_i}) < \infty$. Since $\mathbb{E}_\nu[\bar{\varphi}_{N_i}] \rightarrow 0$, Chebyshev and Borel–Cantelli give $\bar{\varphi}_{N_i} \rightarrow \int \varphi d\mu_{u_c}$ ν -a.e.

For the gaps, let $N_i \leq N < N_{i+1}$. As $\varphi \in \mathcal{B}$ is bounded,

$$|\bar{\varphi}_N - \frac{N_i}{N} \bar{\varphi}_{N_i}| \leq \frac{N_{i+1} - N_i}{N_i} \|\varphi\|_\infty.$$

Because $a < 1$, $\log N_{i+1} - \log N_i = (i+1)^a - i^a = O(i^{a-1}) = o(1)$, so $N_{i+1}/N_i \rightarrow 1$ and likewise $N_i/N \rightarrow 1$. Hence the gap term vanishes and $\bar{\varphi}_N \rightarrow \int \varphi d\mu_{u_c}$ ν -a.e. The two conditions $a\beta > 1$ and $a < 1$ are simultaneously satisfiable iff $\beta > 1$, which is the source of the restriction; for $0 < \beta \leq 1$ the second-moment method gives only the L^2 conclusion of Theorem 1.2.

Since ν was an arbitrary probability measure with density in $\mathcal{B}$, and these are mutually equivalent to Lebesgue on the attractor, the convergence holds Lebesgue-a.e. $\square$

5 Numerical verification

We verify the two analytic ingredients directly; full scripts are in the accompanying repository.

Density stability (Lemma 3.2). Estimating h_u by a 10^7 -step orbit histogram for $u = u_c - \Delta$, $\Delta \in \{10^{-3}, \dots, 5 \times 10^{-2}\}$, and comparing in L^1 to h_{u_c} , a log–log fit gives

$$\|h_u - h_{u_c}\|_{L^1} \sim |u - u_c|^{0.51},$$

consistent with a Hölder modulus $\gamma \approx \frac{1}{2}$ in Lemma 3.2 (Figure 1(a)). We stress that this is a sanity check on the density-stability input, not a verification of the exponent, which our proof leaves as some $\gamma \in (0, 1]$.

Sequential Birkhoff convergence (Theorem 1.2). For $\varphi = \mathbf{1}_{\{x < 0\}}$ and the schedule $u_n = u_c - c(\log n)^{-\beta}$, the time average $\bar{\varphi}_N$ approaches the autonomous value $\int \varphi d\mu_{u_c} \approx 0.2207$. The error decreases with N and the rate increases with β : at $c = 1$, $\beta = 1$, $N = 2 \times 10^7$ the error is 3.8×10^{-3} (Figure 1(b)), consistent with the $\frac{1}{N} \sum_{m \leq N} \Delta_m$ envelope of Proposition 4.1. These are sanity checks on the *mean* convergence of Theorem 1.2; they do not, and cannot, verify the almost-everywhere statement of Corollary 1.3, which concerns a Lebesgue-full set of initial conditions.

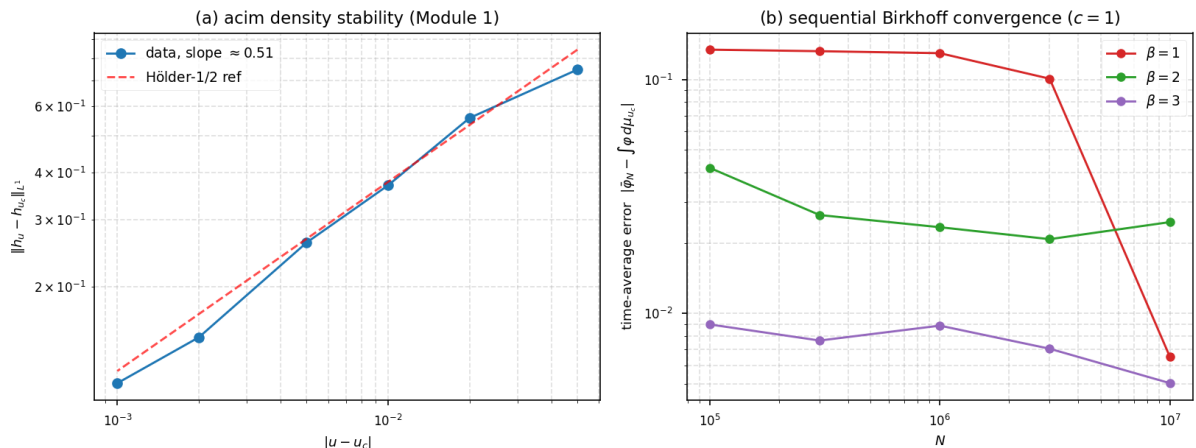


Figure 1: Numerical verification. (a) acim density L^1 -distance vs $|u - u_c|$ on log–log axes, with the Hölder- $\frac{1}{2}$ reference; (b) sequential time-average error vs N for $\beta = 1, 2, 3$ at $c = 1$, showing convergence with β -dependent rate.

6 Concluding remarks

The schedule rate enters only through the tail $\Delta_m = \sup_{k \geq m} |u_k - u_c|$, so the *mean* convergence of Theorem 1.2 holds for every drift with $\Delta_m \rightarrow 0$, including all logarithmic rates $\beta > 0$ and all polynomial rates; the almost-everywhere statement (Corollary 1.3) is established here only for $\beta > 1$, the threshold at which the variance $O(\frac{1}{N} \sum_{m \leq N} \Delta_m)$ becomes summable along a ratio-one subsequence. Closing the gap $0 < \beta \leq 1$ for a.e. convergence would require a maximal inequality or a martingale approximation, which we do not pursue.

Scope of the application. The logarithmic case is highlighted because it is the one arising in non-autonomous symbolic models of arithmetic densities [7], where an aging parameter $u_n = u_c - c(\log n)^{-2}$ is tuned to a $1/\log n$ density envelope. It is important to delimit precisely what the present theorem supplies there. Theorem 1.2 shows that the *Cesàro averages* of bounded observables along the drifting orbit converge to the autonomous u_c -average; equivalently, the symbolic cylinder frequencies of the drifting orbit asymptotically inherit the band-merging statistics μ_{u_c} . This is the ergodic foundation that the symbolic model had assumed rather than proved.

It does *not*, by itself, produce the $1/\log n$ *density envelope* of the raw arithmetic sequence: an ordinary Birkhoff average converges to a constant, whereas the prime-density envelope is a non-stationary, time-decaying quantity. Recovering the envelope requires an additional non-stationary ingredient—a shrinking-target or weighted/thinned estimator, or an externally imposed aging mechanism—which lies outside the scope of this paper. We make this boundary explicit to avoid conflating the two statements: what is proved here is the convergence of stationary cylinder frequencies, not the emergence of a non-stationary density profile.

We expect the method (inducing + uniform Lasota–Yorke + Keller–Liverani + second-moment subsequence argument) to extend to multimodal and higher-dimensional piecewise-expanding families admitting a uniform inducing scheme in the sense of Assumption 1.1 across the relevant parameter set.

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